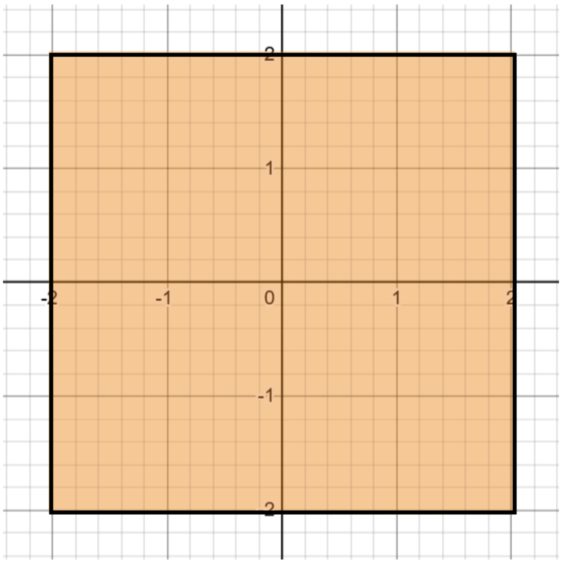
Week 2

1. Set Notations

- $\mathbb{R}$ $\rightarrow$ Set of all real numbers
- $\mathbb{R}_+$ $\rightarrow$ Set of all positive real numbers, including 0
- $\mathbb{Z}$ $\rightarrow$ Set of all integers
- $\mathbb{Z}_+$ $\rightarrow$ Set of all positive integers, including 0
- $[a,b] \;=\; \{x \in \mathbb{R} | \; a \leqslant x \leqslant b\}$
- $(a,b) \;=\; \{x \in \mathbb{R} | \; a < x < b\}$
- $\mathbb{R}^d \;=\; \mathbb{R} \times \mathbb{R} \times \ldots d - times.... \times \mathbb{R}$
- $[a,b]^d \;=\; \left\{x \times \mathbb{R}^d \;\middle|\; x_i \in [a,b],\; i \in \{1,2,\ldots,d\}\right\}$

Example : $[-2,2]^2$



We can also write $[-2, 2]^2$ as a cartesian product : $[-2, 2] \times [-2, -2]$

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2. De Morgan's Law

Let A & B be two sets. Then the following are true

$$(A \cup B)^C = A^C \cap B^C$$

$$(A \cap B)^C = A^C \cup B^C$$

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3. Symbols

For All : ∀ & There exist : ∃

Without Symbol

If V is a vector space, then **for all** $u \in V$, **there exists** a $u' \in V$ such that $u + u' = v_0$, where v_0 is the zero vector of V.

With Symbol

If V is a vector space, then $\forall u \in V,\ \exists u' \in V$ such that $u + u' = v_0$, where v_0 is the zero vector of V.

Implies : $\Longrightarrow$

- $x = 3 \Longrightarrow x^2 = 9$

Equivalent : $\Longleftrightarrow$

- n is divisible by 6 $\Longleftrightarrow$ n is divisibe by 2 and 3

4. Distance & Norm

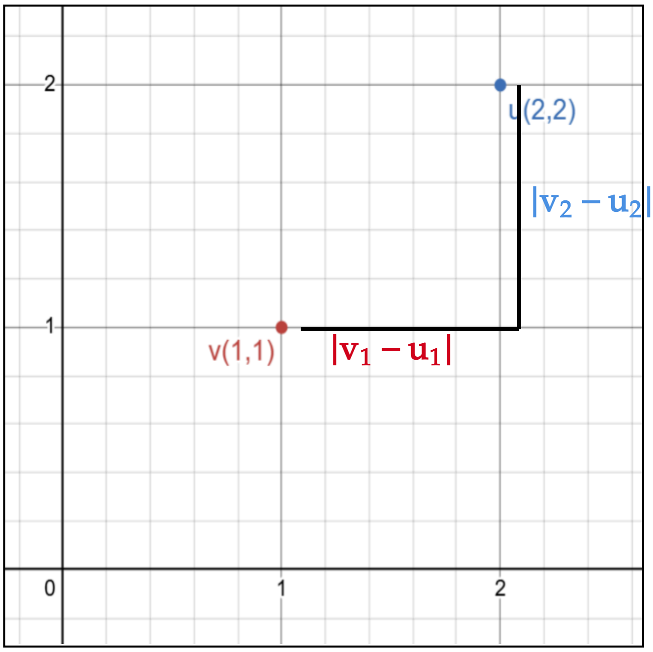
Note : In this section, we will use the word **point** and **vector** interchangably

Let $v, u \in \mathbb{R}^d$ where $v = (v_1, v_2,..., v_d)$ and $u = (u_1, u_2,..., u_d)$.

Manhattan Distance

The Manhattan Distance between two points v and u is given by:

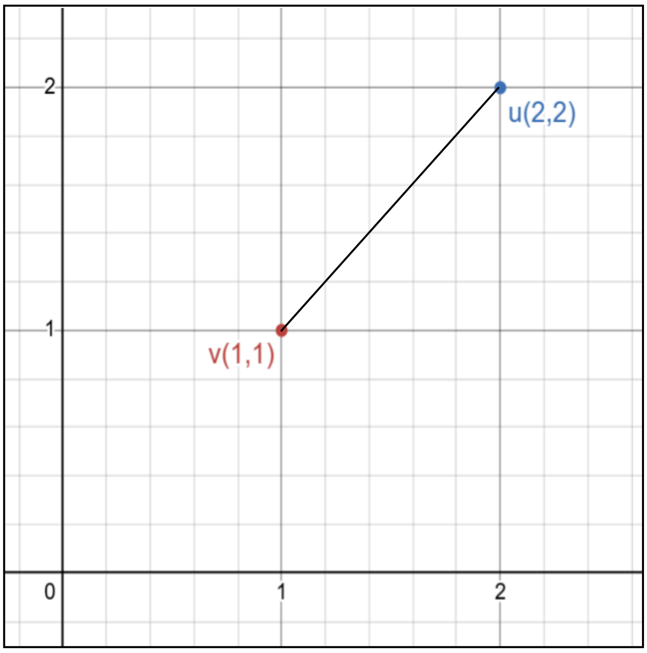
$$\sum_{i=1}^{d} |v_i - u_i| = |v_1 - u_1| + |v_2 - u_2| + ... + |v_d - u_d|$$



Euclidean Distance

The Euclidean Distance between two points u and v is given by

$$\sqrt{\sum_{i=1}^{d} (v_i - u_i)^2} = \sqrt{(v_1 - u_1)^2 + (v_2 - u_2)^2 + ... + (v_d - u_d)^2}$$



l_1 Norm

The l_1 norm (also known as the **Manhattan** norm) of vector v is given by

$$||v||_1 = |v_1| + |v_2| + \dots + |v_d|$$

l_2 Norm

The l_2 norm (also known as the **Euclidean** norm) of vector u is given by

$$||u||_2 = \sqrt{u_1^2 + u_2^2 + \dots + u_d^2} = \sqrt{u^T u}$$

(here, we assume u is a column vector, therefore, u^T will be a row vector)

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5. Metric Space

A metric space is an ordered pair : (M, d), where:

- M is a **set**
- $d : M \times M \rightarrow \mathbb{R}$ is a distance or metric function

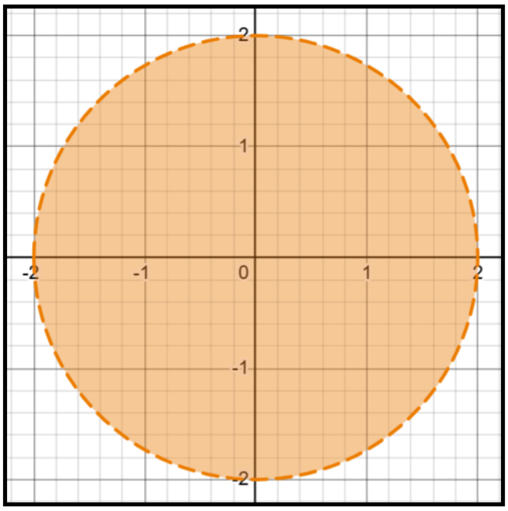
Suppose $x, y \in M$, and $\epsilon \in \mathbb{R}$, we define $B(x, \epsilon)$ and $\overline{B}(x, \epsilon)$ as follows:

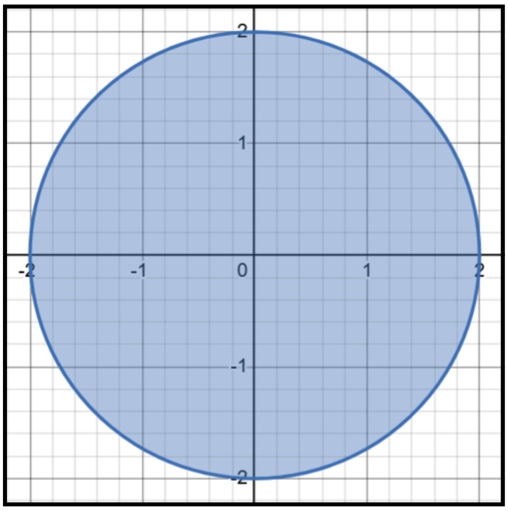
- $B(x, \epsilon) = \left\{ y \middle| d(x, y) < \epsilon \right\}$
- $\overline{B}(x, \epsilon) = \left\{ y \middle| d(x, y) \leqslant \epsilon \right\}$

Example

Let $(\mathbb{R}^2, d)$ be a metric space where,

$$d(x, y) = \|x - y\|_2$$





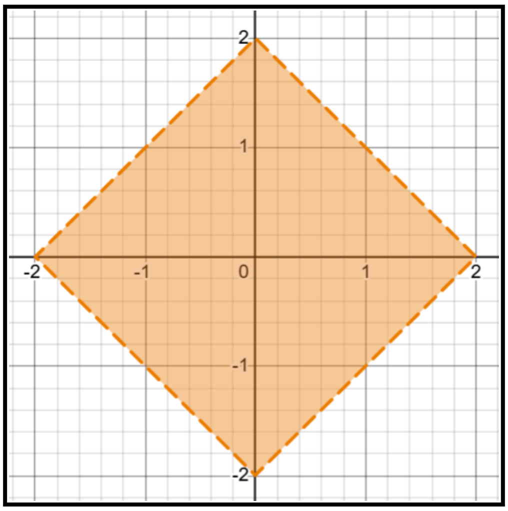
$$B\big((0,0),2\big) = \left\{(x,y) \in R^2 \,\middle|\, \sqrt{x^2+y^2} < 2\right\}$$

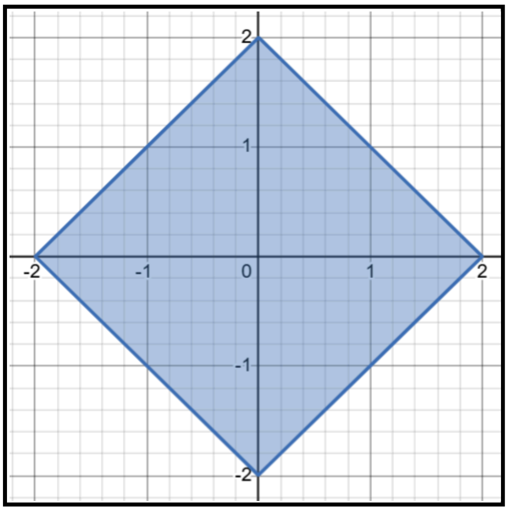
$$\bar{B}\big((0,0),2\big) = \left\{(x,y) \in R^2 \,\middle|\, \sqrt{x^2+y^2} \leqslant 2\right\}$$

Example

Let $(\mathbb{R}^2, d)$ be a metric space where,

$$d(x, y) = \|x - y\|_1$$





$$B\big((0,0),2\big) = \left\{(x,y) \in R^2 \,\middle|\, |x| + |y| < 2\right\}$$

$$\bar{B}\big((0,0),2\big) = \left\{(x,y) \in R^2 \,\middle|\, |x| + |y| \leqslant 2\right\}$$

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6. Sequences

A sequence is a list of numbers in a given order.

$$a_1, a_2, \ldots, a_n, \ldots$$

Sequences can be described by writing rules that specify their terms.

$$a_n = \frac{n-1}{n}$$

or, by listing the terms:

$$\{a_n\} = \left\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots\right\}$$

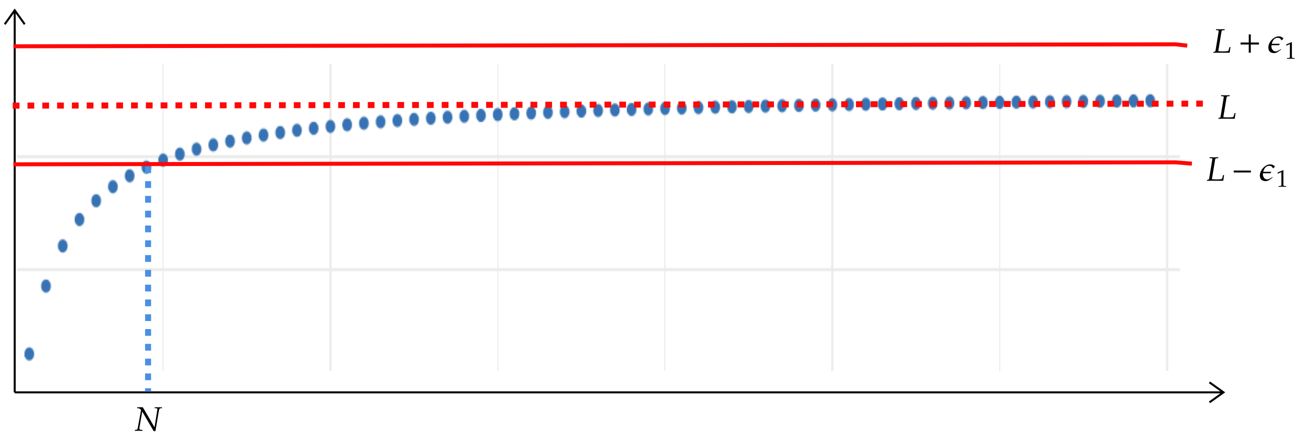
A sequence $\{a_n\}$ **converges** to L if for every positive number ϵ there corresponds an integer N such that

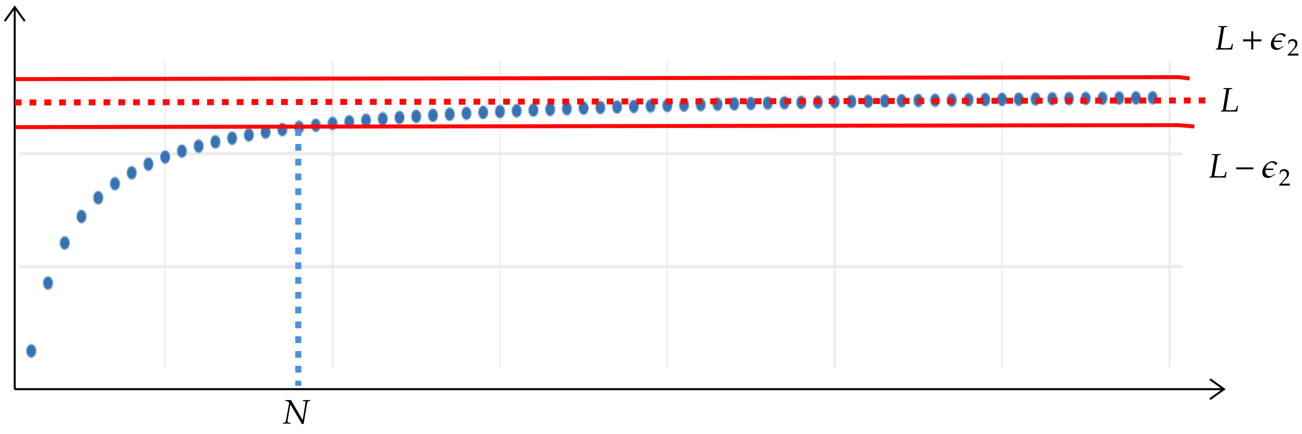
$$|a_n - L| < \epsilon \quad \text{whenever} \quad n \geqslant N$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.

If $\{a_n\}$ converges to L, we write $\lim\limits_{n\to\infty} a_n = L$, or simply $a_n \to L$. and call L the **limit** of the sequence.

Look at the diagram below. We take two epsilons : ϵ_1 and ϵ_2, where $\epsilon_1 > \epsilon_2 > 0$. Similarly, no matter how small epsilon becomes (as long as it is greater than 0), we should be able to find N, such that, for all $n \geqslant N$, $|a_n - L| < \epsilon$.





7. Graph

Consider a function $f:\mathbb{R}^d \to \mathbb{R}$

Graph of f, denoted by G_f is a subset of $\mathbb{R}^{d+1}$.

$$G_f = \left\{(x, f(x)) \middle| x \in \mathbb{R}^d\right\}$$

Suppose $x \in \mathbb{R}^2$, then $(x, f(x)) = (x_1, x_2, f(x))$

8. Continuity and Differentiability

Continuity

A function $f:\mathbb{R} \to \mathbb{R}$ is continuous at $x^* \in \mathbb{R}$, if for all $a_n \to x^*$, $f(a_n) \to f\left(x^*\right)$.

Consider the following function:

$$f(x) = \begin{cases} \frac{|x|}{x}, \; if \; x \neq 0 \\ 0, \; if \; x = 0 \end{cases}$$

Consider the following sequence:

$$a_n = \frac{1}{n}$$

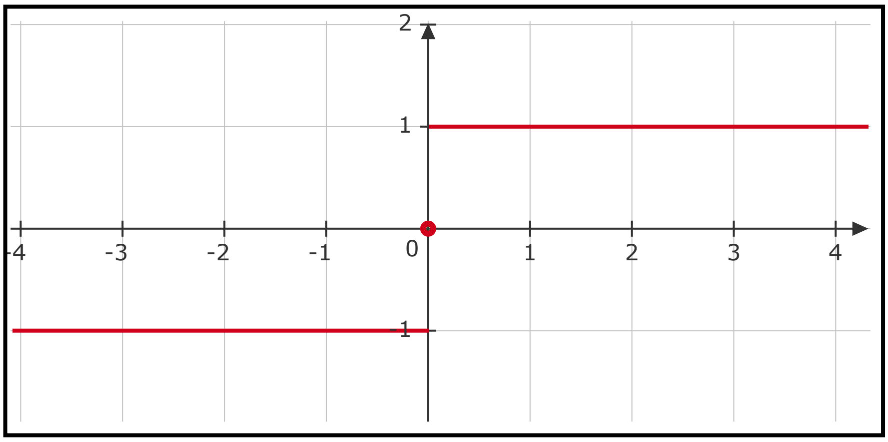
We can see that $a_n \to 0$. However, $f(a_n) \to 1$. Therefore, f is not continuous at 0.

Also, if we consider the following sequence,

$$b_n = -\frac{1}{n}$$

we see that $b_n \to 0$ and $f(b_n) \to -1$.

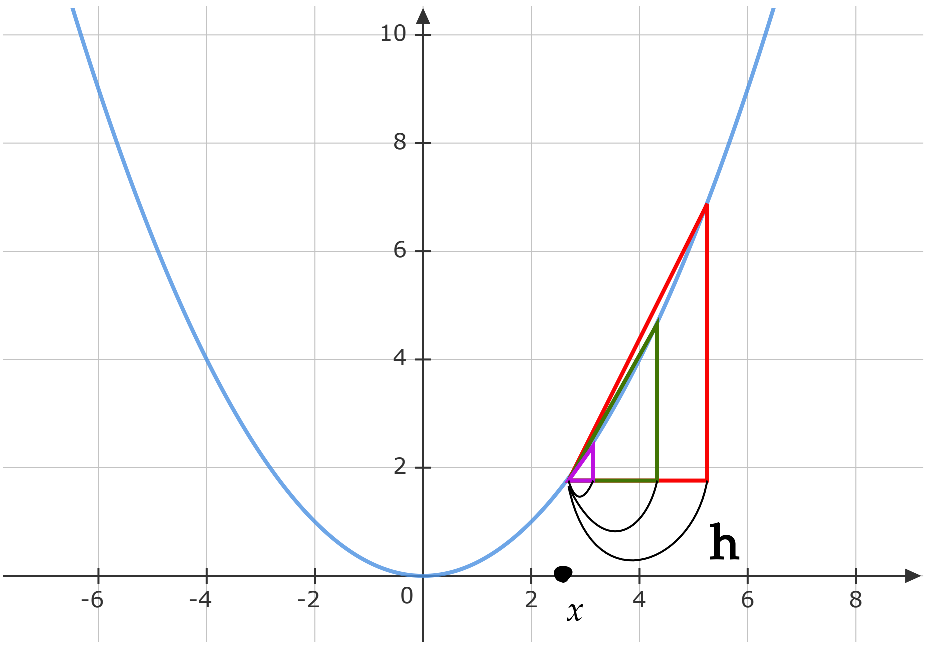
Both a_n and b_n converge to 0. But, $f(a_n) \to 1$ and $f(b_n) \to -1$.



Differentiability

A function $f : \mathbb{R} \to \mathbb{R}$ is differentiable at $x \in \mathbb{R}$, if the following limit exist:

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{(x+h) - (x)}$$



As $h \to 0$, $\lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{(x+h) - (x)}$ gives the instantaneous rate of change of the f at x.

9. Linear Approximation

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function differentiable at $c \in \mathbb{R}$. The linear approximation for f at c is given by:

$$f(x) \approx f(c) + f'(c)(x - c)$$

Informal proof:

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

when x is "close" to c,

$$f'(c) \approx \frac{f(x) - f(c)}{x - c}$$

$$f(x) \approx f(c) + f'(c)(x - c)$$

In simple words, when x is close to c, we can linearly approximate the value of $f(x)$ evaluating $f(c) + f'(c)(x - c)$.

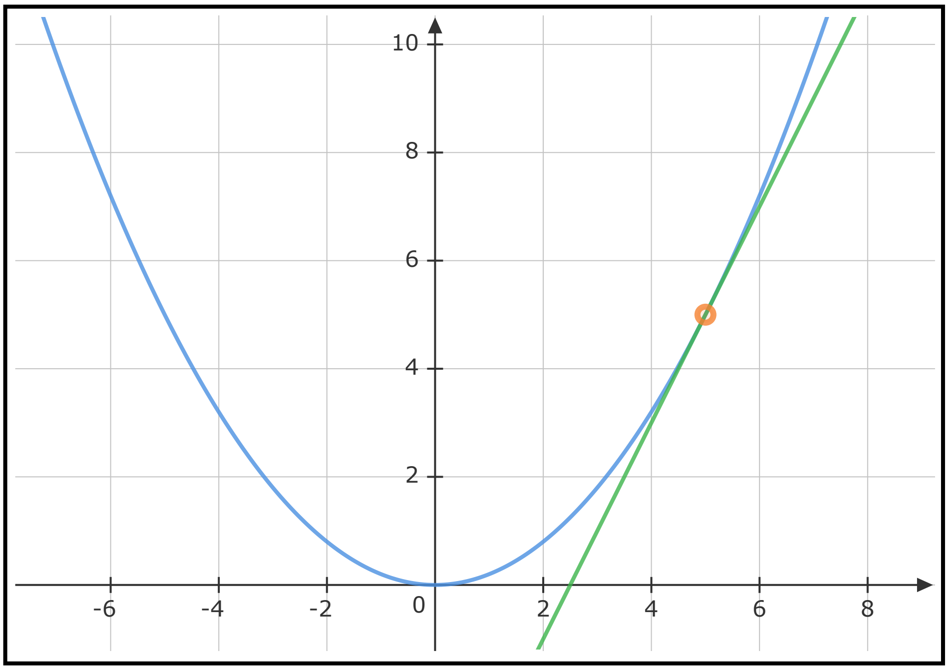
Consider $f(x) = (0.2)x^2$. The linear apporximation for f at 5 is given by:

$$f(x) \approx f(5) + f'(5)(x - 5)$$

$$f(x) \approx (0.2 \times 25) + (0.2 \times 10)(x - 5)$$

$$f(x) \approx 5 + 2x - 10$$

$$f(x) \approx 2x - 5$$



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10. Quadratic Approximation

Let $f : \mathbb{R} \to \mathbb{R}$ be a function differentiable at $x = c$. The quadratic approximation for f at x is given by:

$$f(x) \approx f(c) + f'(c)(x-c) + \frac{1}{2}f''(c)(x-c)^2$$

Consider $f(x) = x^3 - 3x + 1$. The quadratic approximation for f at -1 is given as follows

$$\begin{aligned} f(x) &\approx f(-1) + f'(-1)(x-(-1)) + \frac{1}{2}f''(-1)(x-(-1)^2 \\ &\approx 3 + f'(-1)(x+1) + \frac{1}{2}f''(-1)(x+1)^2 \end{aligned}$$

$$f'(x) = 3x^2 - 3$$

$$f'(-1) = 3 - 3$$

$$f'(-1) = 0$$

$$f''(x) = \frac{d}{dx}(f'(x))$$

$$f''(x) = \frac{d}{dx}\left(3x^2 - 3\right)$$

$$f''(x) = 6x$$

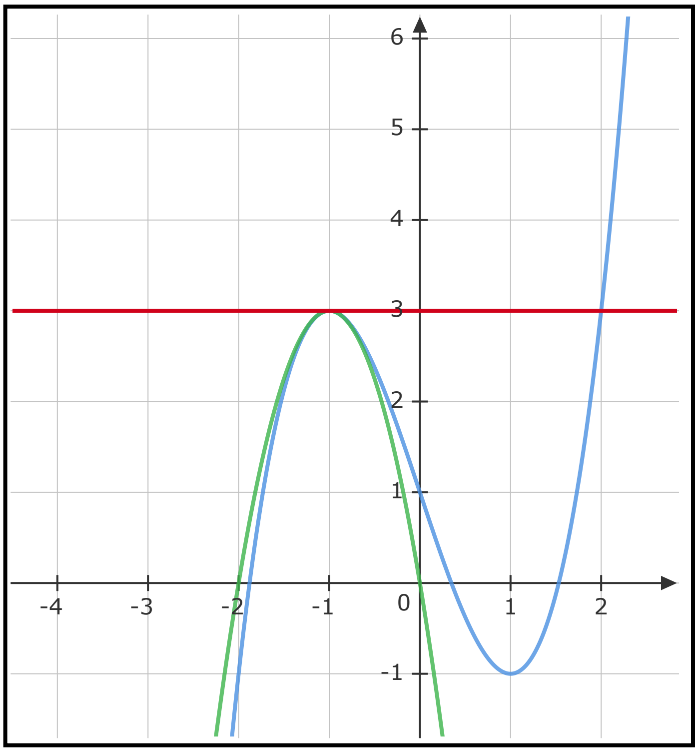
$$f''(-1) = -6$$

$$
\begin{aligned}
f(x) &\approx 3 + 0(x+1) - \left(\frac{1}{2}\right)6\left(x^2 + 2x + 1\right) \\
&\approx 3 - 3x^2 - 6x - 3
\end{aligned}
$$

Let us also find the linear approximation of f at -1.

$$
\begin{aligned}
f(x) &\approx f(-1) + f'(-1)(x - (-1)) \\
&\approx 3 + 0(x+1) \\
&\approx 3
\end{aligned}
$$

Here, the blue curve is $f(x)$, red line is its linear apporximation at -1 and green is its quadratic approximation at -1.



We can see that the quadratic approximation is better than the linear approximation.

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11. Product Rule

Let $f(x) = g(x) \cdot h(x)$. Assume f is differentiable at c.

$$
f'(c) = g'(c) \cdot h(c) + h'(c) \cdot g(c)
$$

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12. Chain Rule

Let $f(x) = p\big(q(x)\big)$. Assume:

- p is differentiable at $q(x)$
- q is differentiable at x

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{p\big(q(x+h)\big) - p\big(q(x)\big)}{h} \end{aligned}$$

Multiply and divide by $q(x+h) - q(x)$

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \left[\frac{p\big(q(x+h)\big) - p\big(q(x)\big)}{q(x+h) - q(x)} \cdot \frac{q(x+h) - q(x)}{h} \right] \\ &= \lim_{h \to 0} \frac{p\big(q(x+h)\big) - p\big(q(x)\big)}{q(x+h) - q(x)} \cdot \lim_{h \to 0} \frac{q(x+h) - q(x)}{h} \end{aligned}$$

As $h \to 0$, $q(x+h) - q(x) \to 0$

$$\lim_{h \to 0} \frac{p\big(q(x+h)\big) - p\big(q(x)\big)}{q(x+h) - q(x)} = p'\big(q(x)\big)$$

Also,

$$\lim_{h \to 0} \frac{q(x+h) - q(x)}{h} = q'(x)$$

Together,

$$f'(x) = p'\big(q(x)\big) \cdot q'(x)$$

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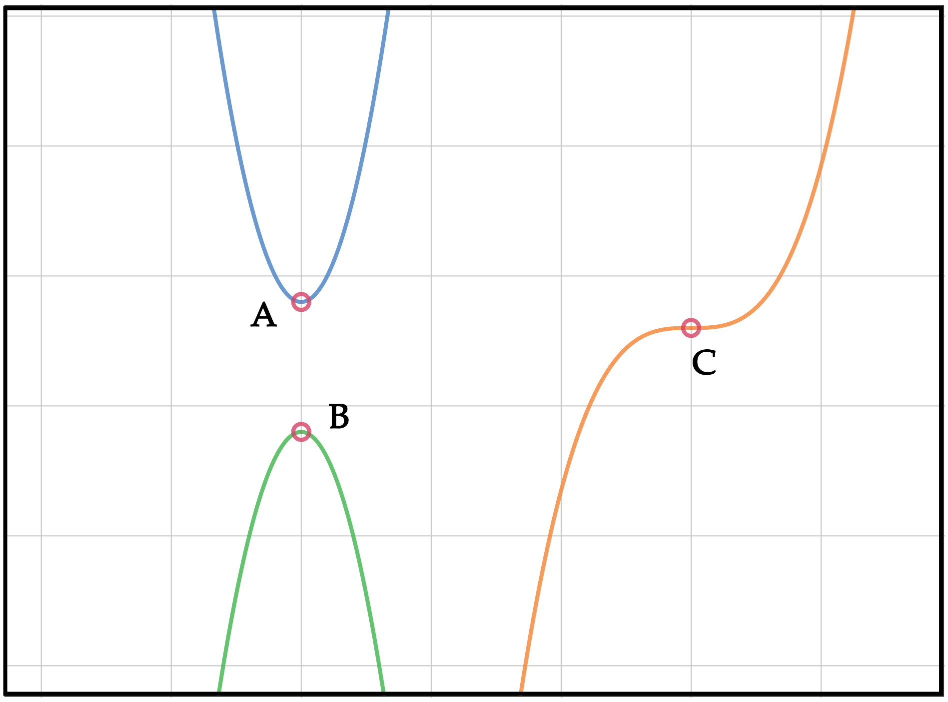
13. Local Maxima, Local Minima, Saddle Point

Consider $f : \mathbb{R} \to \mathbb{R}$

- c is a **local minima** of f if there exists a real number $\alpha > 0$, such that for all $x \in (c - \alpha, c + \alpha)$, $f(c) \leqslant f(x)$
- c is a **local maxima** of f if there exists a real number $\alpha > 0$, such that for all $x \in (c - \alpha, c + \alpha)$, $f(c) \geqslant f(x)$
- c is a **critical point** if $f'(c) = 0$ or $f'(c)$ does not exist
- All local minimum and local maximum are critical points
- A critical point which is neither a local minima nor a local maxima is a **saddle point**



- $A \longrightarrow$ Local Minima
- $B \longrightarrow$ Local Maxima
- $C \longrightarrow$ Saddle point

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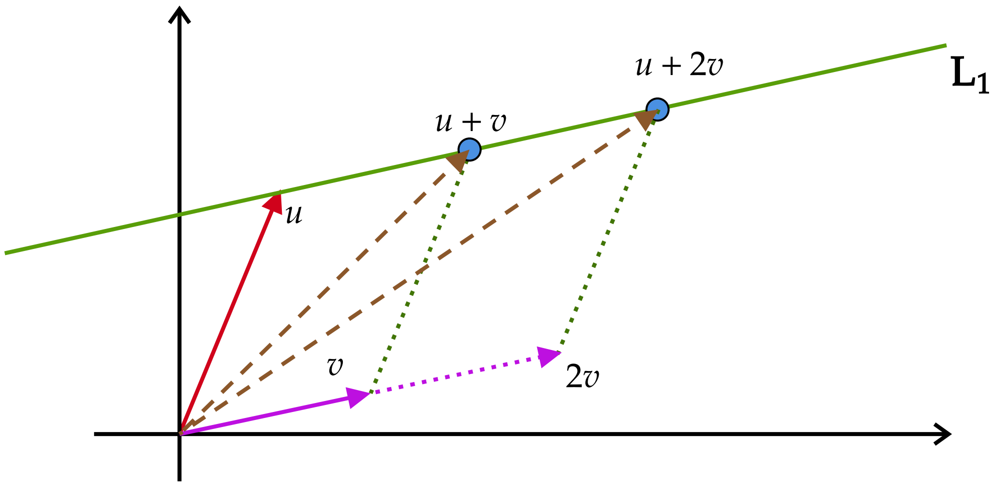
14. Geometry of Lines

Consider $u, v \in \mathbb{R}^d$.

A line through u, along the direction of v is given by

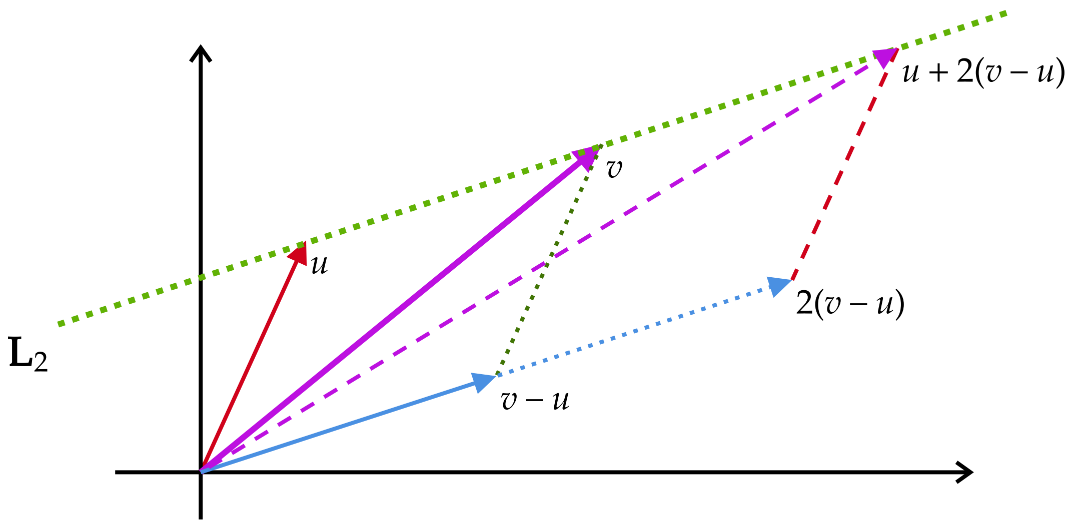
$$L_1 = \left\{ x \in \mathbb{R}^d \middle| x = u + kv,\ k \in \mathbb{R} \right\}$$

The intuition for the above set can be seen in the diagram below. We can clearly see that the line L follows **parallelogram law of addition.**



A line through u and v is given by the following set:

$$L_2 = \left\{ x \in \mathbb{R}^d \,\middle|\, x = u + k \cdot (v-u),\ k \in \mathbb{R} \right\}$$



Hyperplane

Let $b \in \mathbb{R}$.

Given a vector $w \in \mathbb{R}^d$, we can describe a hyperplane which is normal to w as follows:

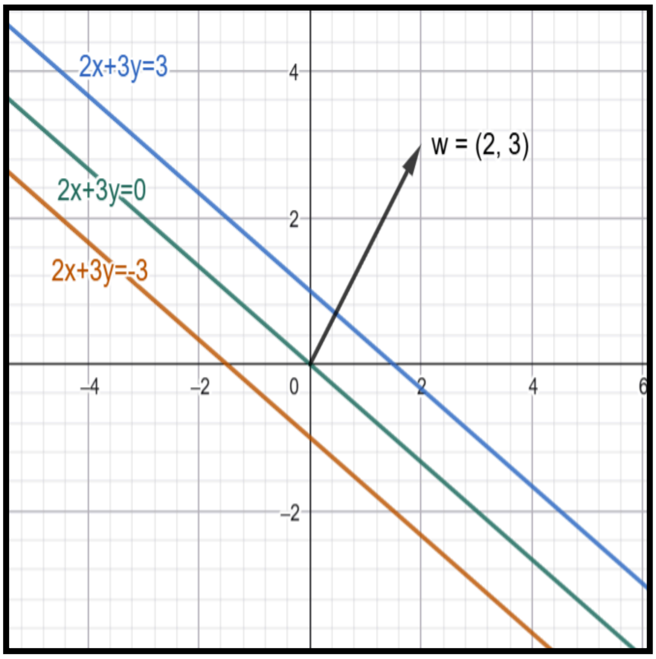
$$\left\{ x \in \mathbb{R}^d \,\middle|\, w^T x = b \right\}$$

A hyperplane in $\mathbb{R}^d$ is has dimension $d-1$.

Example in $\mathbb{R}^2$

Let $w = (2, 3)$. The hyperplane normal to w is given by

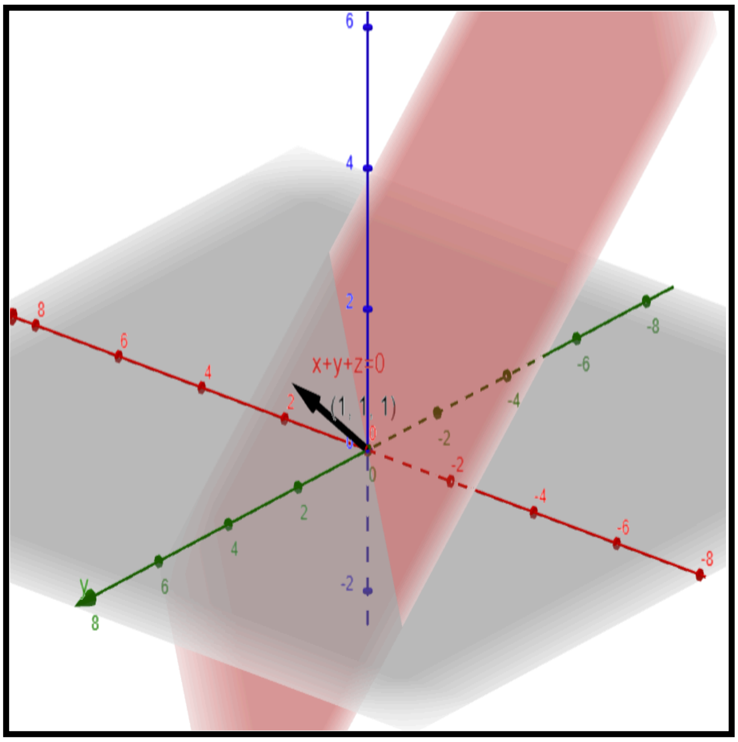
$$\left\{(x,y) \in \mathbb{R}^2 \middle| 2x + 3y = b\right\}$$



Example in $\mathbb{R}^3$

Let $w = (1,1,1)$, $k = (x,y,z)$. The hyperplane normal to w is given by

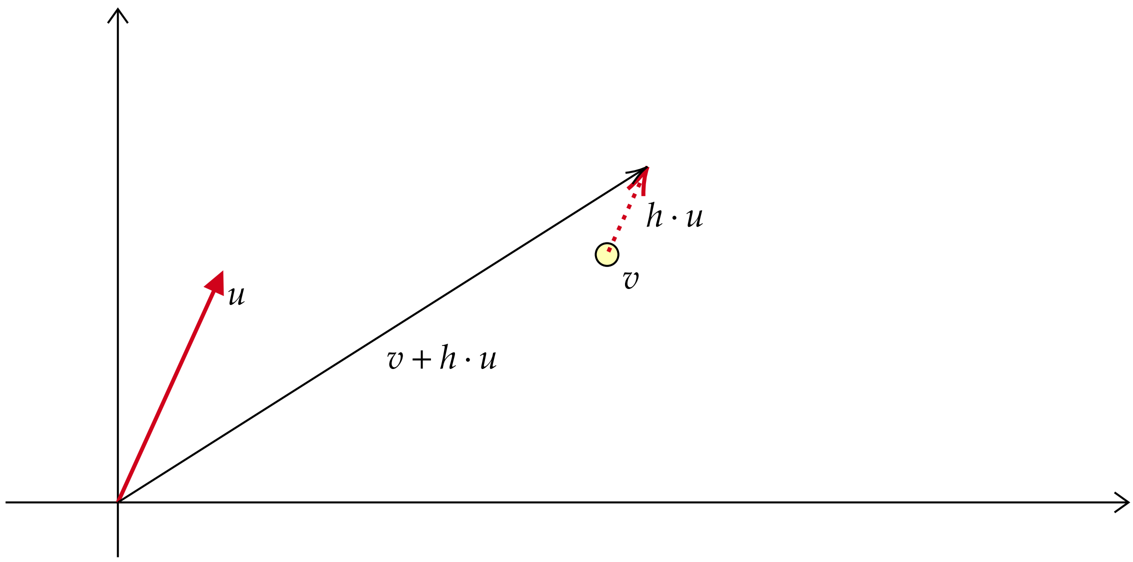
$$\left\{(x,y,z) \in \mathbb{R}^3 \middle| x + y + z = b\right\}$$



15. Directional Derivative, Partial Derivative, Gradient

Directional Derivative

Consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Let $v, u \in \mathbb{R}^2$ and let $h \in \mathbb{R}$.
Here, u is a unit vector. That is, $||u|| = 1$



From v, if we happen to move h units in the direction of u, which is a unit vector, the function's value at the new point is $f(v + hu)$.

At v, the instantaneous rate of change of the function f in the direction of u is given by:

$$D_u f(v) = \lim_{h \rightarrow 0} \frac{f(v + hu) - f(v)}{h}$$

The above limit, if it exists, is known as the directional derivative of f at v in the direction of the unit vector u.

Partial Derivative

In $\mathbb{R}_2$, if u happens to be:

- $(1, 0)$, we call it the **partial derivative** with respect to x_1
- $(0, 1)$, we call it the **partial derivative** with respect to x_2

In $\mathbb{R}^n$, let $x_i = (0, 0, ..., 1 ...., 0)$, where 1 is in the i^{th} index, where $i \in \{1, 2, ..., n\}$.

The directional derivative in the direction of x_i can also be called as the partial derivative with respect to x_i. It is given by

$$\left.\frac{\delta f}{\delta x_i}\right|_v = D_{x_i} f(v)$$

A partial derivative measures how the function changes when we move only along one coordinate direction, keeping all other coordinates fixed.

Gradient

Let $f : \mathbb{R}^n \to \mathbb{R}$. Let $v, u \in \mathbb{R}^n$ and let $||u|| = 1$.

The **gradient** of f at v is given by

$$\nabla f(v) = \left(\left.\frac{\delta f}{\delta x_1}\right|_v, \cdots, \left.\frac{\delta f}{\delta x_n}\right|_v \right)$$

In simple words, gradient vector is the collection of parital derivatives.

We can compute the directional derivative of f at v in the direction of a unit vector u as follows

$$D_u f(v) = \nabla f(v) \cdot u$$

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16. Direction of Steepest Ascent, Descent and No Change

Cosine function's value at $0°, 90°$ & $180°$

$$\cos(0°) = 1 \qquad \cos(90°) = 0 \qquad \cos(180°) = -1$$

Note, the maximum value of the cosine function is 1 and the minimum value of the cosine function is -1.

Angle between two unit vectors

Let $p, q \in \mathbb{R}^n$. Suppose, the angle between p and q is θ°. Then,

$$\frac{p}{||p||} \cdot \frac{q}{||q||} = \cos(\theta°)$$

$$=$$

$$p \cdot q = ||p|| \times ||q|| \times \cos(\theta°)$$

Few questions

Consider a function $f : \mathbb{R}^n \to \mathbb{R}$. Let $u, v \in \mathbb{R}^n$ and let $||u|| = 1$.

We know that the directional derivative of f at v in the direction of the unit vector u is given by

$$D_u f(v) = \nabla f(v) \cdot u$$

$$D_u f(v) = ||\nabla f(v)|| \times ||u|| \times \cos(\alpha°)$$

$$D_u f(v) = ||\nabla f(v)|| \times \cos(\alpha°)$$

Here, $\alpha°$ is the angle between $\nabla f(v)$ and u.

We can ask a few questions here:

1. At the point v, in which direction does the function increase most rapidly?
2. At the point v, in which direction does the function decrease most rapidly?
3. At the point v , in which direction does the function remain unchanged?

At the point v, in which direction does the function increase most rapidly?

Here, we are essentially asking the following:

- At the point v, along which direction, is the directional derivative maximum?

$$D_u f(v) = ||\nabla f(v)|| \times \cos(\alpha°)$$

For the directional derivative to be maximum, we would want $\cos(\alpha°)$ to be maximum. This happens when $\alpha° = 0°$.

Therefore, when $u = \dfrac{\nabla f(v)}{||\nabla f(v)||}$, the directional derivative at v is maximum.

This direction is also called the direction of steepest ascent.

At the point v, in which direction does the function decrease most rapidly?

Here, we are essentially asking the following:

- At the point v, along which direction, is the directional derivative minimum?

$$D_u f(v) = ||\nabla f(v)|| \times \cos(\alpha^\circ)$$

For the directional derivative to be minimum, we would want $\cos(\alpha^\circ)$ to be minimum. This happens when $\alpha^\circ = 180^\circ$.

Therefore, when $u = \dfrac{-\nabla f(v)}{||\nabla f(v)||}$, the directional derivative at v is minimum.

This direction is also called the direction of steepest descent.

At the point v , in which direction does the function remain unchanged?

Here, we are essentially asking the following:

- At the point v, along which direction, is the directional derivative 0?

$$D_u f(v) = ||\nabla f(v)|| \times \cos(\alpha^\circ)$$

For the directional derivative to be 0, we would want $\cos\left(\alpha^\circ\right)$ to be maximum. This happens when $\alpha^\circ = 90^\circ$.

Therefore, when $\nabla f(v)^T u = 0$, the directional derivative at v is 0.

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17. Linear Approximation (Multivariable Functions)

Let $f : \mathbb{R}^n \to \mathbb{R}$. Let $v \in \mathbb{R}^n$. The linear approximation of $f$at v is given by:

$$f(x) \approx f(v) + \nabla f(v) \cdot (x - v)$$

Example

Let $f(x, y) = x^2 + y^2 + 4$.

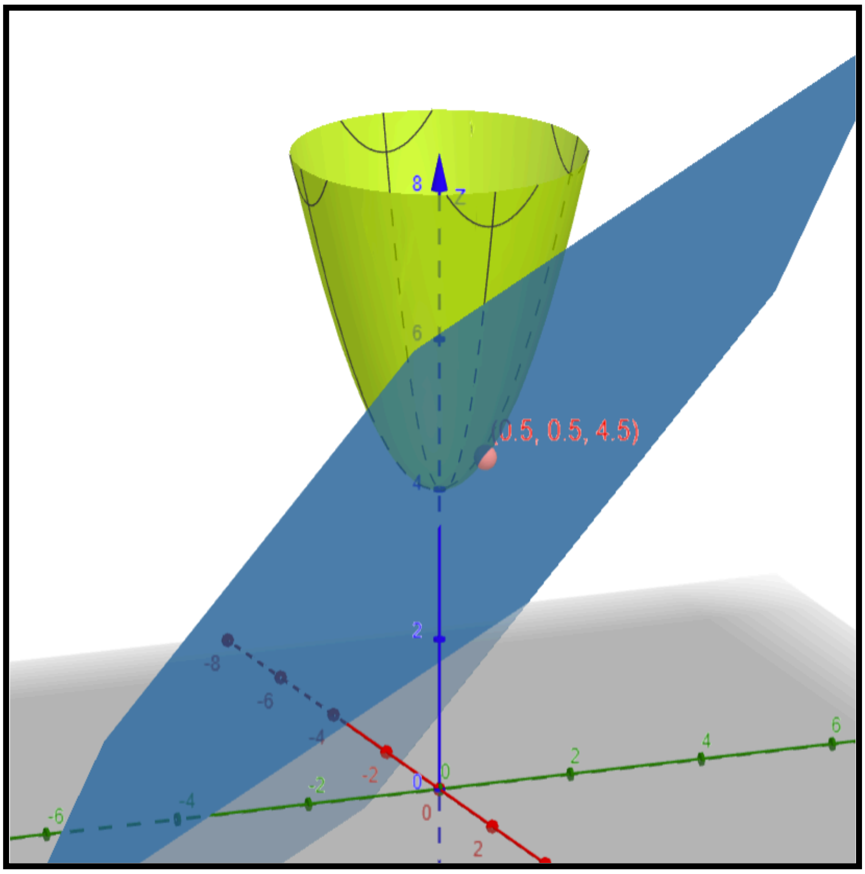
Let us find the linear approximation of f at $\left(\dfrac{1}{2}, \dfrac{1}{2}\right)$

The gradient of f is given by

$$\nabla f(x,y) = \left(\frac{\delta f}{\delta x}, \frac{\delta f}{\delta y}\right)$$
$$= (2x, 2y)$$

The gratient of f at $\left(\frac{1}{2}, \frac{1}{2}\right)$ is $(1,1)$.

$$f(x,y) \approx f\left(\left(\frac{1}{2}, \frac{1}{2}\right)\right) + \nabla f\left(\left(\frac{1}{2}, \frac{1}{2}\right)\right) \cdot \left(x - \left(\frac{1}{2}, \frac{1}{2}\right)\right)$$
$$\approx \frac{1}{4} + \frac{1}{4} + 4 + (1,1) \cdot \left(\frac{2x-1}{2}, \frac{2y-1}{2}\right)$$
$$\approx \frac{18}{4} + \frac{2x-1}{2} + \frac{2y-1}{2}$$
$$\approx \frac{4x - 2 + 4y - 2 + 18}{4}$$
$$\approx \frac{4x + 4y + 14}{4}$$



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18. Cauchy-Schwarz Inequality

We know the following

$$-1 \leqslant \cos(\theta°) \leqslant 1$$

Let $a, b \in \mathbb{R}^n$ and let $\alpha°$ be the angle between a and b. Then,

$$\frac{a^T b}{||a|| \cdot ||b||} = \cos(\alpha°)$$

Therefore,

$$-1 \leqslant \frac{a^T b}{||a|| \cdot ||b||} \leqslant 1$$

$$\Downarrow$$

$$-||a|| \cdot ||b|| \leqslant a^T b \leqslant ||a|| \cdot ||b||$$

This inequality is also known as the **Cauchy-Schwarz** inequality.

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19. Triangular Inequality

$$\begin{aligned} ||a+b||^2 &= (a+b)^T(a+b) \\ &= a^T a + 2a^T b + b^T b \\ &= ||a||^2 + 2a^T b + ||b||^2 \end{aligned}$$

Rearanging terms:

$$\frac{||a+b||^2 - ||a||^2 - ||b||^2}{2} = a^T b$$

We know that $a^T b \leqslant ||a|| \cdot ||b||$

$$\frac{||a+b||^2 - ||a||^2 - ||b||^2}{2} \leqslant ||a|| \cdot ||b||$$

$$||a+b||^2 \leqslant ||a||^2 + 2 \cdot ||a|| \cdot ||b|| + ||b||^2$$

$$||a+b||^2 \leqslant (||a|| + ||b||)^2$$

Both sides are non-negative. Therefore,

$$\sqrt{||a+b||^2} \leqslant \sqrt{(||a|| + ||b||)^2}$$

$$||a+b|| \leqslant ||a|| + ||b||$$

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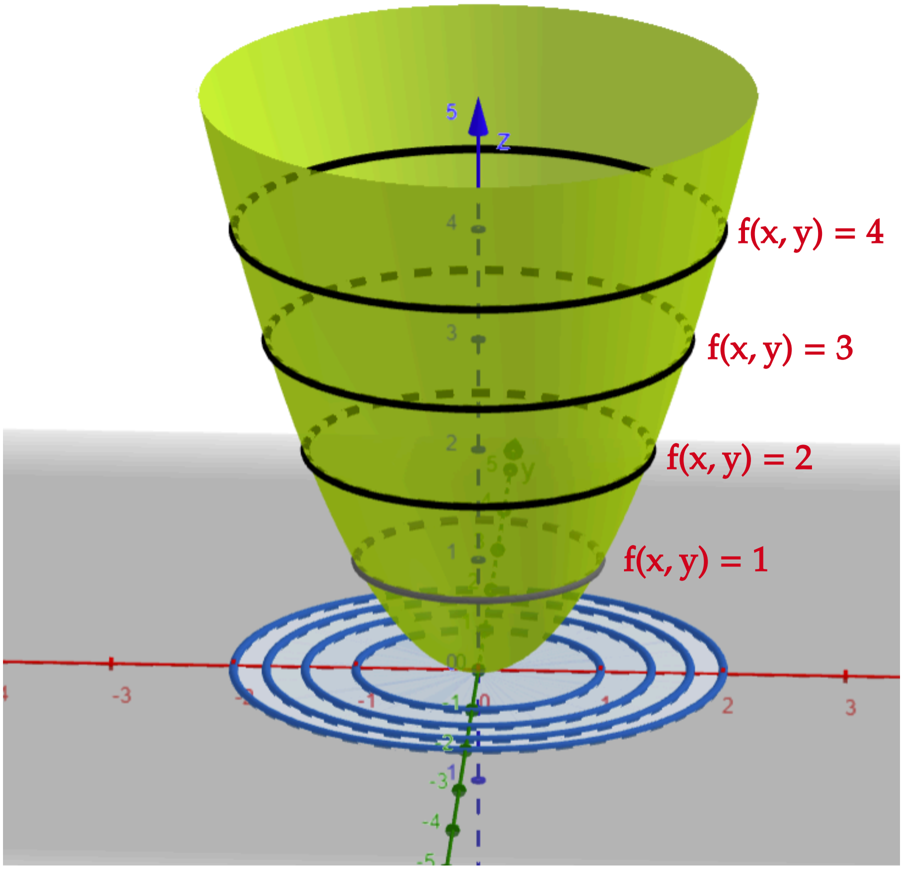
20. Level Curves and Contour Plots

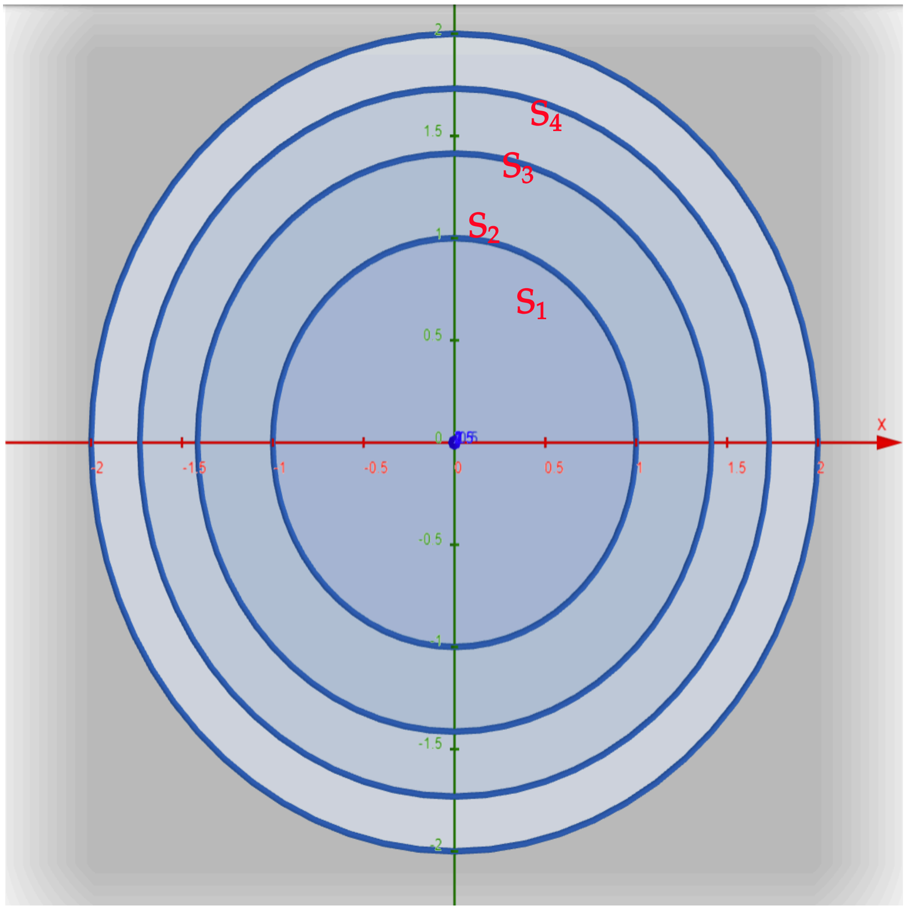
Consider a function $f : \mathbb{R}^2 \to \mathbb{R}$.

For a constant $c \in \mathbb{R}$, the level curve corresponding to c is the set

$$\left\{(x, y) \middle| f(x, y) = c\right\}$$



- $S_1 = \left\{(x, y) \middle| f(x, y) = 1\right\}$
- $S_2 = \left\{(x, y) \middle| f(x, y) = 2\right\}$
- $S_3 = \left\{(x, y) \middle| f(x, y) = 3\right\}$
- $S_4 = \left\{(x, y) \middle| f(x, y) = 4\right\}$



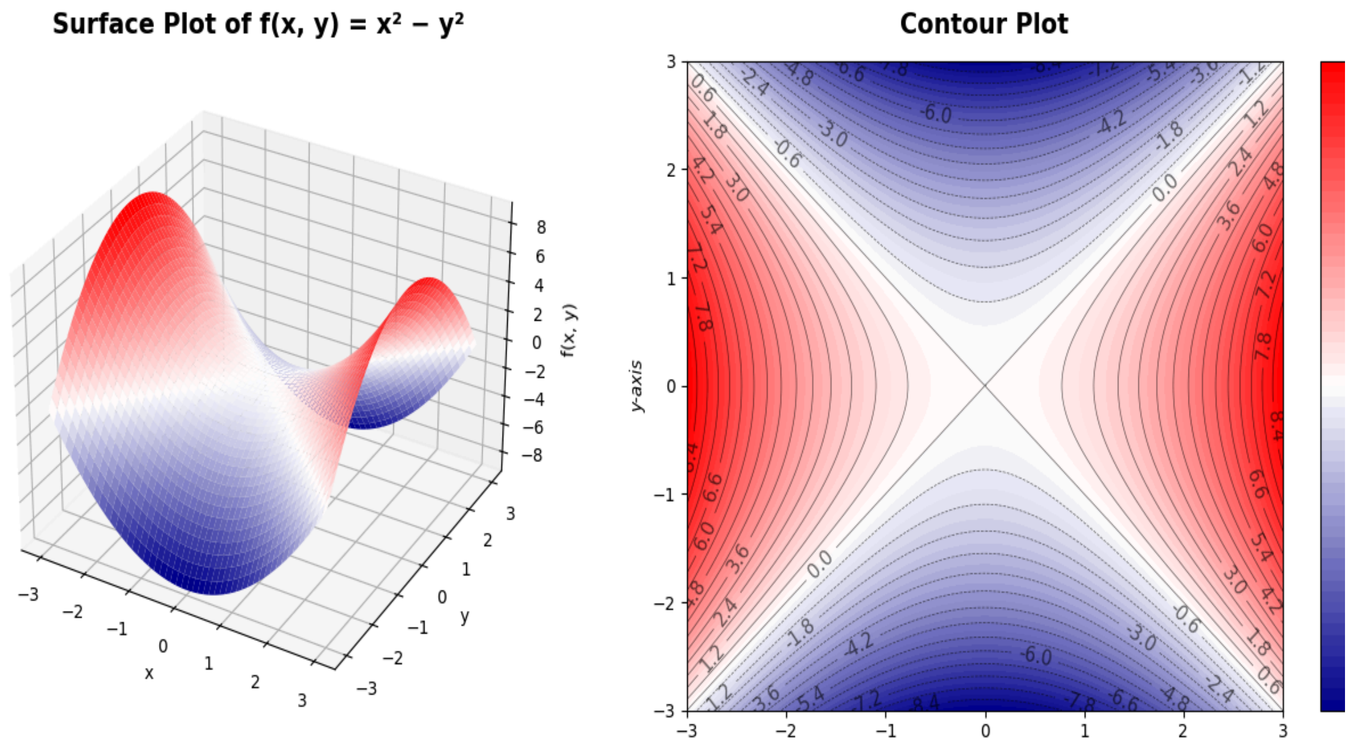
CONTOUR PLOT

In the picture below,

- red region → function's value is positive
- blue region → function's value is negative
- white region → function's value is zero

In the right side, we have the contour plot. Note that in some regions, the distance the adjacent level curves is comparatively small when compared other regions.

- If the distance between adjacent level curves is small in a region, then in that region the function is changing rapidly.
- If the distance between adjacent level curves is large in a region, then in that region the function is changing slowly.

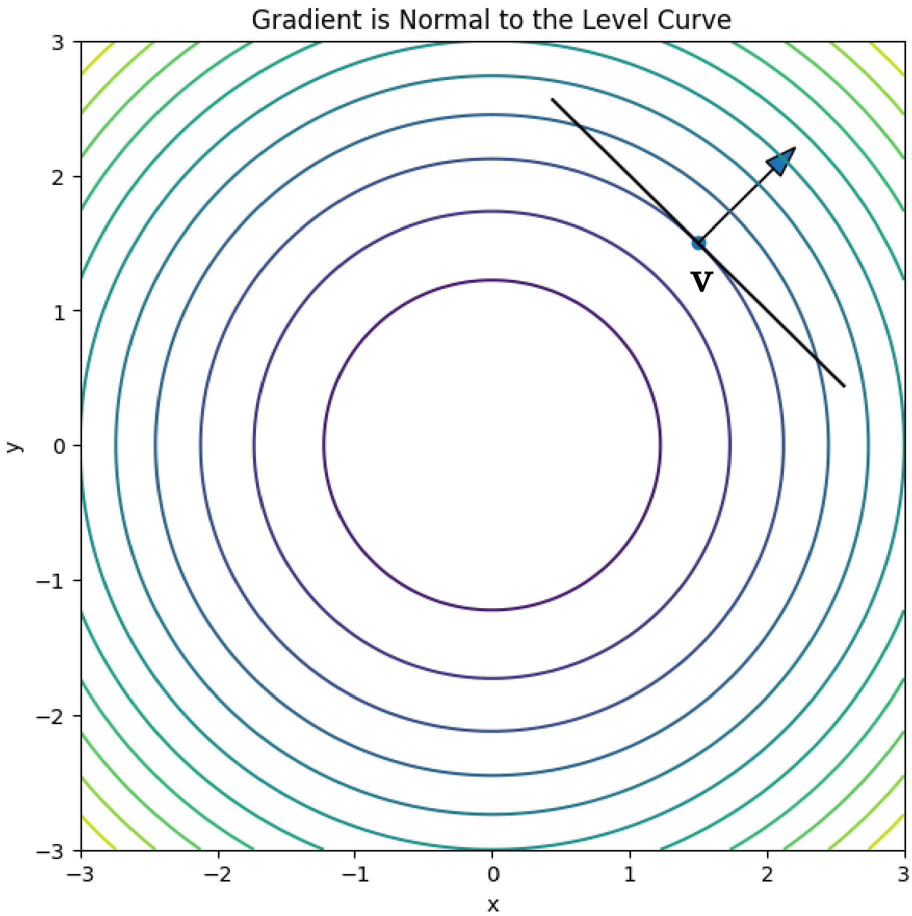


The direction of the gradient vector at a point is normal to the tangent of the level curve at that point.

This is because the gradient vector points in the direction in which the directional derivative at that point is maximum. On the other hand, the tangent direction to the level curve is a direction in which the function does not change. Therefore, the directional derivative in the tangent direction is zero.

In the picture below, please note that it is not the gradient vector. It just represents the **direction** of the gradient vector at v.

We can see that the gradient's direction is perpendicular to the tangent of the level curve at v



Gradient is Normal to the Level Curve
v
x
y